Com S 4720/5720 Fall 2025

Problem Set 7 (55 pts)

Due at 11:59pm

Friday, October 24

All the problems are from *https://aimacode.github.io/aima-exercises/* (possibly) with minor typo fixes, slight rewriting, or additions of missing information.

1.(6 pts) (Exercise 7.5) Which of the following are correct? No explanation is needed. The operator $\models$ always has the lowest precedence.

(a) False $\models$ True.

(c) $A \land B \models A \Leftrightarrow B$.

(e) $A \Leftrightarrow B \models \neg A \lor B$.

(h) $(A \lor B) \land \neg(A \Rightarrow B)$ is satisfiable.

(j) $(C \lor (\neg A \land \neg B)) \equiv ((A \Rightarrow C) \land (B \Rightarrow C))$

(k) $(A \Leftrightarrow B) \land (\neg A \lor B)$ is satisfiable.

2.(4 pts) (Exercise 7.7(c)) Prove, or find a counterexample to the following assertion:

$$\text{If } \alpha \models (\beta \lor \gamma) \text{ then } \alpha \models \beta \text{ or } \alpha \models \gamma \text{ (or both).}$$

3.(15 pts) (Exercise 7.23) Consider the following sentence:

$$\Big((\text{Food} \Rightarrow \text{Party}) \lor (\text{Drinks} \Rightarrow \text{Party})\Big) \Rightarrow \Big((\text{Food} \land \text{Drinks}) \Rightarrow \text{Party}\Big).$$

(a) (5 pts) Determine, using enumeration, whether this sentence is valid, satisfiable (but not valid), or unsatisfiable.

(b) (5 pts) Convert the left-hand and right-hand sides of the main implication into CNF, showing each step, and explain how the results confirm your answer to (a).

(c) (5 pts) Prove your answer to (a) using resolution.

4.(6 pts) (Exercise 8.11) Consider a vocabulary with the following symbols:

- *Occupation*(p, o): Predicate. Person p has occupation o.
- *Customer*(p_1, p_2): Predicate. Person p_1 is a customer of person p_2.
- *Boss*(p_1, p_2): Predicate. Person p_1 is a boss of person p_2.
- *Doctor*, *Surgeon*, *Lawyer*, *Actor*: Constants denoting occupations.
- *Emily*, *Joe*: Constants denoting people.

Use these symbols to write the following assertions in first-order logic:

(c) (2 pts) All surgeons are doctors.

(e) (2 pts) Emily has a boss who is a lawyer.

(f) (2 pts) There exists a lawyer all of whose customers are doctors.

5.(12 pts) (Exercise 8.29) For each of the following sentences in English, decide if the accompanying first-order logic sentence is a good translation. If not, explain why not and correct it.

(a) (4 pts) Any apartment in London has lower rent than some apartments in Paris.

$$\forall x \ \Big(\mathit{Apt}(x) \land \mathit{In}(x, \mathit{London})\Big) \Rightarrow \exists y \ \Big((\mathit{Apt}(y) \land \mathit{In}(y, \mathit{Paris})) \Rightarrow (\mathit{Rent}(x) < \mathit{Rent}(y))\Big).$$

(b) (4 pts) There is exactly one apartment in Paris with rent below \$1000.

$$\exists x \ \mathit{Apt}(x) \land \mathit{In}(x, \mathit{Paris}) \land \forall y \ \Big(\mathit{Apt}(y) \land \mathit{In}(y, \mathit{Paris}) \land (\mathit{Rent}(y) < \mathit{Dollars}(1000))\Big) \Rightarrow (y = x).$$

(c) (4 pts) If an apartment is more expensive than all apartments in London, it must be in Moscow.

$$\forall x \ \mathit{Apt}(x) \land \Big(\forall y \ \mathit{Apt}(y) \land \mathit{In}(y, \mathit{London}) \land (\mathit{Rent}(x) > \mathit{Rent}(y))\Big) \Rightarrow \mathit{In}(x, \mathit{Moscow}).$$

6.(12 pts) (Exercise 8.36) Consider a first-order logical knowledge base that describes worlds containing people, songs, albums (e.g., "Meet the Beatles") and disks (i.e., particular physical instances of CDs). The vocabulary contains the following symbols:

- *CopyOf*(d, a): Predicate. Disk d is a copy of album a.
- *Owns*(p, d): Predicate. Person p owns disk d.
- *Sings*(p, s, a): Album a includes a recording of song s sung by person p.
- *Wrote*(p, s): Person p wrote song s.
- *Lennon*, *McCartney*, *Gershwin*, *Joe*, *ADayInTheLife*, *Revolver*

Use the above vocabulary to write the following sentences in first-order logic:

(c) (3 pts) Either Lennon or McCartney wrote “A Day in the Life”.

(f) (3 pts) Every song that McCartney sings on *Revolver* was written by McCartney.

(h) (3 pts) Every song that Gershwin wrote has been recorded on some album. (Possibly different songs are recorded on different albums.)

(i) (3 pts) There is a single album that contains every song that Joe has written.